Dorey's Resource Book

Topic 3 – Right-angled triangles

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1 Pythagoras' theorem

ANCHOR QUESTION

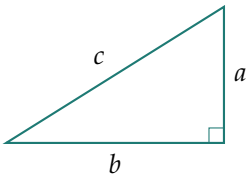
A right-angled triangle has shorter sides 9 and 12. Find the length of the hypotenuse.

SYLLABUS MST-12-S1-04

- Apply Pythagoras' theorem to solve practical problems in two dimensions

NOTES

- The **hypotenuse** is the longest side of a right-angled triangle. It is opposite the right angle.
- **Pythagoras' theorem:** $c^2 = a^2 + b^2$, where c is the hypotenuse.
- To find the hypotenuse, add the squares: $c = \sqrt{a^2 + b^2}$.
- To find a shorter side, subtract the squares: $a = \sqrt{c^2 - b^2}$.
- Write all sides in the same units first. Round only at the end.



WORKED EXAMPLE

A right-angled triangle has shorter sides 8 and 15. Find the length of the hypotenuse, x.

$x^2 = 8^2 + 15^2$ hypotenuse: add the squares

$x^2 = 64 + 225 = 289$

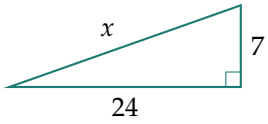
$x = \sqrt{289} = 17$ square root

Tier 1 · Finding the hypotenuse

1. The shorter sides of a right-angled triangle are 3 and 4. Find the hypotenuse.

2. The shorter sides of a right-angled triangle are 5 and 12. Find the hypotenuse.

3. Find x.



Tier 2 · Rounding

4. The shorter sides of a right-angled triangle are 4 and 9. Find the hypotenuse, correct to two decimal places.

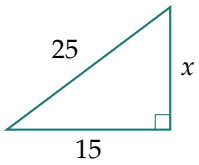
5. The shorter sides of a right-angled triangle are 2.5 and 6. Find the hypotenuse.

6. The shorter sides of a right-angled triangle are 11 and 14. Find the hypotenuse, correct to one decimal place.

Tier 3 · Finding a shorter side

7. A right-angled triangle has a hypotenuse of 10 and one shorter side of 8. Find the other shorter side.

8. Find x.



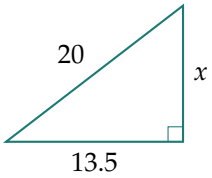
9. A right-angled triangle has a hypotenuse of 17 and one shorter side of 8. Find the other shorter side.

Tier 4 · Shorter side with rounding

10. A right-angled triangle has a hypotenuse of 12 and one shorter side of 7. Find the other shorter side, correct to two decimal places.

11. A right-angled triangle has a hypotenuse of 8.5 and one shorter side of 4.2. Find the other shorter side, correct to two decimal places.

12. Find x, correct to two decimal places.



Tier 5 · Mixed units

13. The shorter sides of a right-angled triangle are 60 cm and 0.8 m. Find the hypotenuse in metres.

14. A right-angled triangle has a hypotenuse of 2.6 m and a shorter side of 100 cm. Find the other side in metres.

15. The shorter sides of a right-angled triangle are 1.5 km and 800 m. Find the hypotenuse in kilometres.

Tier 6 · Two triangles

16. In triangle ABC, D is a point on BC and AD is perpendicular to BC. $AB = 13$, $BD = 5$ and $DC = 16$. Find AD, then find AC.

17. In triangle ABC, AD is perpendicular to BC. $AB = 10$, $BD = 6$ and $DC = 6$. Find AC.

18. A square has sides of 7 cm. Find the length of its diagonal, correct to two decimal places.

Conceptual

19. Use Pythagoras' theorem to explain why the hypotenuse is always the longest side.

20. A triangle has a hypotenuse of 10 and a shorter side of 6. Mia finds the other side as $\sqrt{100 + 36} = 11.66$. Explain her mistake and find the correct answer.

21. A square has a diagonal of 10 cm. Find the side length, correct to two decimal places.

22. Explain why the sides must be in the same units before using Pythagoras' theorem.

23. A composite shape is made of a right-angled triangle with shorter sides 9 cm and 12 cm, and a semicircle drawn on the hypotenuse.

(a) Find the diameter of the semicircle. (2 marks)

(b) Find the perimeter of the composite shape, to the nearest centimetre. (2 marks)

Revision

24. A spinner with 5 equal sections numbered 1 to 5 is spun and a coin is tossed.

(a) How many outcomes are there? (1 mark)

(b) Find $P(\text{head and an even number})$. (1 mark)

(c) Find the probability of NOT getting a head and an even number. (1 mark)

(d) Find the expected number of these results in 300 goes. (2 marks)

25. A garden bed is a trapezium with parallel sides of 6 m and 9 m, which are 4 m apart.

(a) Find the area of the garden bed. (2 marks)

(b) Mulch is spread 0.1 m deep. Find the volume of mulch needed. (1 mark)

(c) Mulch costs \$65 per m^3, plus a \$45 delivery fee. Find the total cost. (2 marks)

26. A tap fills a 9000 L tank at a rate of 15 L/min.

(a) How many hours does it take to fill the tank? (2 marks)

(b) Convert 15 L/min to L/h. (1 mark)

(c) Water costs \$3.20 per kilolitre. Find the cost of the water to fill the tank. (2 marks)

2 Pythagoras in practical problems

ANCHOR QUESTION

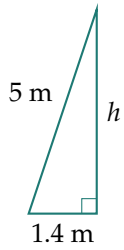
A 5 m ladder leans against a wall with its foot 1.4 *m from the wall. How high up the wall does it reach?*

SYLLABUS MST-12-S1-04

- Apply Pythagoras' theorem to solve practical problems in two dimensions
- Solve practical problems involving Pythagoras' theorem, the trigonometry of right-angled triangles, angles of elevation and depression, and true bearings and compass bearings

NOTES

- Draw a right-angled triangle for the situation and label the lengths you know.
- The hypotenuse is the side opposite the right angle: the ladder, the diagonal, the kite string, the sloping ramp.
- Walking due north then due east (or any two perpendicular directions) makes a right angle.
- Answer the question asked: a saving or a difference may need one more step after Pythagoras.
- State units and round as asked.



WORKED EXAMPLE

A 6 m ladder leans against a wall. Its foot is 2 m from the wall. How high up the wall does it reach, correct to two decimal places?

The ladder is the hypotenuse. — opposite the right angle
$h^2 = 6^2 - 2^2 = 32$ — shorter side: subtract
$h = \sqrt{32} = 5.656\ldots \approx 5.66$ m — round at the end

Tier 1 · Ladders and walls

27. A 3.5 m ladder leans against a wall with its foot 1 m from the wall. How high up the wall does it reach, correct to two decimal places?

28. A ladder reaches 4 m up a wall. Its foot is 1.2 m from the wall. How long is the ladder, correct to two decimal places?

29. A 10 m ladder reaches 8 m up a wall. How far is its foot from the wall?

Tier 2 · Diagonals

30. A rectangular phone screen is 15 cm by 7 cm. Find its diagonal, correct to one decimal place.

31. A gate is 1.8 m wide and 1.2 m high. A diagonal brace is fitted across it. Find the length of the brace, correct to two decimal places.

32. A TV is sold as having a 65-inch diagonal. Its width is 57 inches. Find its height, correct to the nearest inch.

Tier 3 · Directions

33. Ella walks 600 m due south, then 800 m due west. How far is she from her start, in a straight line?

34. A boat sails 12 km due east, then 5 km due north. How far is it from its starting point?

35. A plane flies 230 km due north, then 95 km due east. How far is it from its starting point, correct to the nearest kilometre?

Tier 4 · Comparing distances

36. A rectangular field is 120 m by 50 m. How much shorter is the walk along the diagonal than along two sides?

37. A square courtyard has sides of 20 m. How much shorter is it to cut diagonally across than to walk around two sides, correct to one decimal place?

38. A ramp rises 0.5 m over a horizontal distance of 6 m. The ramp surface is to be covered with non-slip strip costing \$18 per metre. Find the cost, to the nearest dollar.

Conceptual

39. A ladder is moved further from the wall. Explain what happens to the height it reaches.

40. Sam walks 3 km north, then 4 km north-east. He says he is 5 km from his start. Explain why Pythagoras' theorem does not apply.

41. A 4 m ladder must reach 3.9 m up a wall. How far from the wall can its foot be, correct to two decimal places?

42. Explain why the diagonal of a rectangle is always shorter than the sum of its length and width.

43. A plan of a triangular paddock ABC has a scale of $1 : 400$. The angle at B is a right angle.

(a) On the plan, AB measures 15 cm. Find the actual length of AB in metres. (2 marks)

(b) BC is the same length as AB. Find AC, correct to 3 significant figures. (3 marks)

Revision

44. A bag holds 4 green, 6 yellow and 10 blue marbles. One marble is taken at random.

(a) Find $P(\text{yellow})$ in simplest form. (1 mark)

(b) Find the probability that the marble is not blue, as a percentage. (2 marks)

(c) In 60 draws with replacement, how many greens would you expect? (2 marks)

45. A cylindrical can has a diameter of 8 cm and a height of 12 cm.

(a) Find the volume of the can, correct to the nearest cm^3. (2 marks)

(b) Find its capacity in millilitres. (1 mark)

(c) Find the area of the curved surface, to the nearest cm^2. (2 marks)

46. A builder's plan has a scale of 1 : 50. A wall measures 9.6 cm on the plan.

(a) Find the actual length of the wall in metres. (2 marks)

(b) The wall is 2.4 m high. Find its area. (1 mark)

(c) 1 L of paint covers 12 m^2. How many 4 L tins for two coats? (2 marks)

3 The trigonometric ratios

ANCHOR QUESTION

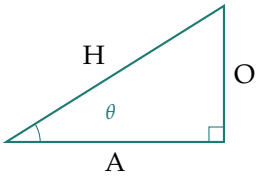
A right-angled triangle has sides 5, 12 and 13. The side opposite θ is 5. Write $\sin\theta$ as a fraction.

SYLLABUS MST-12-S1-04

- Apply trigonometric ratios to find the lengths of unknown sides in right-angled triangles given an angle in both degrees, and degrees and minutes

NOTES

- Name the sides from the marked angle θ. **Hypotenuse (H):** opposite the right angle. **Opposite (O):** across from θ. **Adjacent (A):** next to θ, not the hypotenuse.
- If the marked angle moves to the other acute angle, O and A swap. H never changes.
- $\sin\theta = \dfrac{O}{H}$, $\cos\theta = \dfrac{A}{H}$, $\tan\theta = \dfrac{O}{A}$ (SOH CAH TOA).
- Choose the ratio that uses the two sides in the question: the side you know and the side you want.
- Set the calculator to **degree** mode before finding sin, cos or tan of an angle.



WORKED EXAMPLE

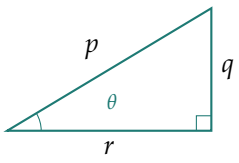
A right-angled triangle has sides 8, 15 and 17. The side opposite θ is 8. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

$\text{H} = 17, \text{O} = 8, \text{A} = 15$ — name the sides from θ

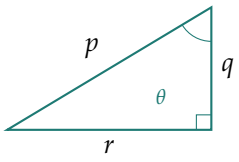
$\sin\theta = \frac{8}{17}$, $\cos\theta = \frac{15}{17}$, $\tan\theta = \frac{8}{15}$ — SOH CAH TOA

Tier 1 · Naming the sides

47. Name the hypotenuse, opposite and adjacent sides for the angle θ.



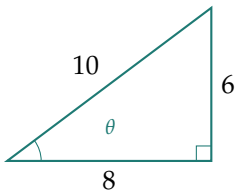
48. Name the hypotenuse, opposite and adjacent sides for the angle θ.



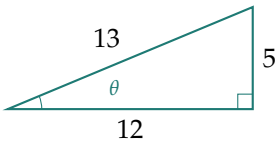
49. In triangle XYZ, the right angle is at Y. For the angle at X, which side is the opposite side?

Tier 2 · Writing one ratio

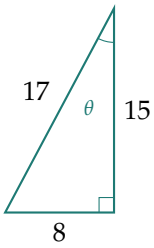
50. Write $\tan\theta$ as a fraction.



51. Write $\cos\theta$ as a fraction.



52. Write $\sin\theta$ as a fraction.



Tier 3 · All three ratios

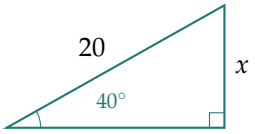
53. A right-angled triangle has sides 3, 4 and 5. The side opposite θ is 3. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

54. A right-angled triangle has sides 7, 24 and 25. The side adjacent to θ is 7. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

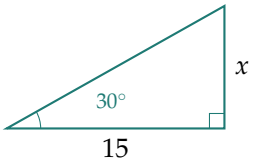
55. A right-angled triangle has sides 20, 21 and 29. The side opposite θ is 20. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

Tier 4 · Choosing the ratio

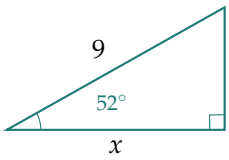
56. Write an equation connecting x and the given side.



57. Write an equation connecting x and the given side.



58. Write an equation connecting x and the given side.



Tier 5 · Using a calculator

59. Find $\sin 50°$, correct to 4 decimal places.

60. Find $\cos 72°$, correct to 4 decimal places.

61. Find $\tan 25°$ and $\tan 65°$, correct to 4 decimal places. Which is greater than 1?

Conceptual

62. Explain why $\sin\theta$ and $\cos\theta$ are always less than 1 for an acute angle in a right-angled triangle.

63. Jake writes $\tan\theta = \frac{A}{O}$. Explain his mistake.

64. In a right-angled triangle, $\sin\theta = \frac{3}{5}$. Find $\cos\theta$ and $\tan\theta$.

65. Explain why the opposite and adjacent sides change when the marked angle moves to the other acute angle.

66. A right-angled triangle has a hypotenuse of 41 and a shorter side of 9, which is opposite the angle θ.

(a) Find the length of the third side. (2 marks)

(b) Write $\tan\theta$ as a fraction. (1 mark)

(c) Which trigonometric ratio of θ is equal to $\frac{40}{41}$? (1 mark)

Revision

67. A ladder 7 m long leans against a wall. Its foot is 1.8 m from the base of the wall.

(a) How high up the wall does the ladder reach, to two decimal places? (2 marks)

(b) Is the height at least 4 times the distance from the wall? Justify. (2 marks)

(c) Write the ratio of distance from the wall to height, in the form $1 : n$. (1 mark)

68. A coin is tossed and a spinner with sections A, B and C is spun.

(a) Draw a tree diagram and list the outcomes. (2 marks)

(b) Find $P(\text{tail and C})$. (1 mark)

(c) Find $P(\text{head or B, or both})$. (2 marks)

69. Priya earns \$1240 a week. Her tax is \$214 a week and the Medicare levy is 2% of her pay.

(a) Find the Medicare levy each week. (1 mark)

(b) Find her net (take-home) pay each week. (2 marks)

(c) Find her net pay for a year (52 weeks). (2 marks)

4 Finding a side: unknown on top

ANCHOR QUESTION

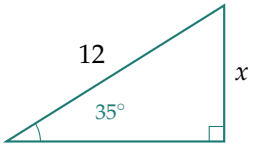
A right-angled triangle has a hypotenuse of 12 and an angle of 35°. Find the side opposite the angle, correct to two decimal places.

SYLLABUS MST-12-S1-04

- Apply trigonometric ratios to find the lengths of unknown sides in right-angled triangles given an angle in both degrees, and degrees and minutes

NOTES

- Label the sides H, O and A from the given angle, then choose the ratio that uses the known side and the unknown side.
- Write the equation, e.g. $\sin 35° = \dfrac{x}{12}$.
- When x is on top, multiply both sides by the bottom number: $x = 12 \sin 35°$.
- Check: the hypotenuse must be the longest side.



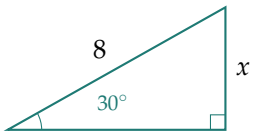
WORKED EXAMPLE

A right-angled triangle has a hypotenuse of 20 and an angle of 25°. Find the side adjacent to the angle, correct to two decimal places.

A and H: cosine — choose the ratio
$\cos 25° = \frac{x}{20}$ — CAH
$x = 20 \cos 25° = 18.126\ldots \approx 18.13$ — multiply by 20

Tier 1 · Sine

70. Find x, correct to two decimal places.

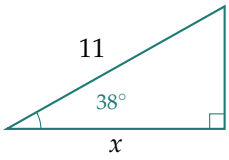


71. A right-angled triangle has a hypotenuse of 15 and an angle of 42°. Find the side opposite the angle, to two decimal places.

72. A right-angled triangle has a hypotenuse of 6.4 and an angle of 67°. Find the side opposite the angle, to two decimal places.

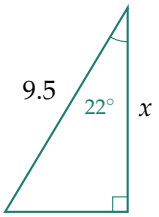
Tier 2 · Cosine

73. Find x, correct to two decimal places.



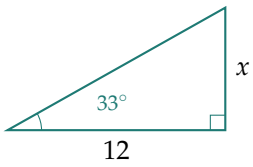
74. A right-angled triangle has a hypotenuse of 24 and an angle of 55°. Find the side adjacent to the angle, to two decimal places.

75. Find x, correct to two decimal places.



Tier 3 · Tangent

76. Find x, correct to two decimal places.

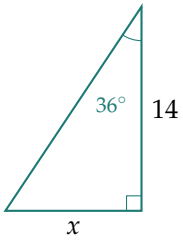


77. A right-angled triangle has a side of 5 adjacent to an angle of 61°. Find the side opposite the angle, to two decimal places.

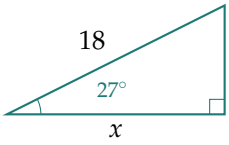
78. A right-angled triangle has a side of 40 adjacent to an angle of 15°. Find the side opposite the angle, to one decimal place.

Tier 4 · Choosing the ratio

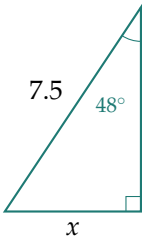
79. Find x, correct to two decimal places.



80. Find x, correct to two decimal places.



81. Find x, correct to two decimal places.



Tier 5 · Practical problems

82. A 7 m ladder leans against a wall, making an angle of 75° with the ground. How high up the wall does it reach, to two decimal places?

83. A kite string is 50 m long and makes an angle of 40° with the ground. How high is the kite, to the nearest metre?

84. A tree casts a shadow 18 m long. The sun's rays make an angle of 34° with the ground. How tall is the tree, to one decimal place?

Conceptual

85. Explain how you decide whether to use sin, cos or tan to find a side.

86. Zoe finds a shorter side as $x = 10\sin 30^\circ = -9.88$. Explain what has gone wrong.

87. In a right-angled triangle with hypotenuse 10, explain why the side opposite 60° is longer than the side opposite 30°.

88. Explain why $\tan 45^\circ = 1$ using a right-angled triangle.

89. A ramp 6.2 m long makes an angle of 6° with the ground.

(a) Find the height of the ramp, to two decimal places. (2 marks)

(b) Find the horizontal distance the ramp covers, to two decimal places. (2 marks)

Revision

90. A rectangular sports field is 105 m long and 68 m wide.

(a) Find the length of the diagonal, to the nearest metre. (2 marks)

(b) Find the area of the field in hectares. (2 marks)

(c) A player jogs around the edge 3 times. How far is this in km? (1 mark)

91. A survey found that 18 of 75 customers bought a coffee with breakfast.

(a) Find the relative frequency of buying a coffee, as a decimal. (1 mark)

(b) Estimate how many of next month's 1200 customers will buy a coffee. (2 marks)

(c) Coffee costs \$5.20. Estimate the coffee sales for the month. (2 marks)

92. A flight leaves Perth at 11:20 pm and lands in Sydney 4 hours 10 minutes later. Sydney is 3 hours ahead of Perth.

(a) What is the time in Perth when the flight lands? (1 mark)

(b) What is the local time in Sydney when the flight lands? (2 marks)

(c) The flight is 3290 km. Find the average speed, to the nearest km/h. (2 marks)

5 Finding a side: unknown on the bottom

ANCHOR QUESTION

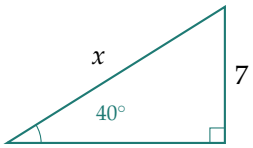
A right-angled triangle has a side of 7 opposite an angle of 40°. Find the hypotenuse, correct to two decimal places.

SYLLABUS MST-12-S1-04

- Apply trigonometric ratios to find the lengths of unknown sides in right-angled triangles given an angle in both degrees, and degrees and minutes

NOTES

- Choose the ratio as before. Sometimes the unknown ends up on the **bottom** of the fraction: $\sin 40° = \dfrac{7}{x}$.
- When x is on the bottom, swap x and the ratio, so you **divide**: $x = \dfrac{7}{\sin 40°}$.
- When x is on top, multiply. Decide which before you press any calculator keys.
- Check: if x is the hypotenuse, it must be longer than the other sides.



WORKED EXAMPLE

A right-angled triangle has a side of 12 adjacent to an angle of 35°. Find the hypotenuse, correct to two decimal places.

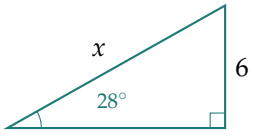
A and H: cosine — choose the ratio

$\cos 35° = \frac{12}{x}$ — x is on the bottom

$x = \frac{12}{\cos 35°} = 14.649\ldots \approx 14.65$ — divide

Tier 1 · Sine

93. Find x, correct to two decimal places.

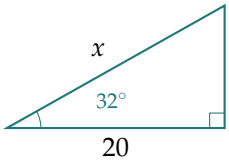


94. A right-angled triangle has a side of 15 opposite an angle of 62°. Find the hypotenuse, to two decimal places.

95. A right-angled triangle has a side of 3.6 opposite an angle of 19°. Find the hypotenuse, to two decimal places.

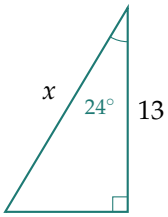
Tier 2 · Cosine

96. Find x, correct to two decimal places.



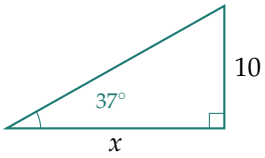
97. A right-angled triangle has a side of 8 adjacent to an angle of 71°. Find the hypotenuse, to two decimal places.

98. Find x, correct to one decimal place.



Tier 3 · Tangent

99. Find x, correct to two decimal places.



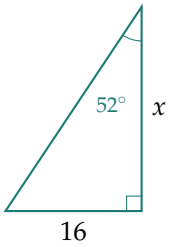
100. A right-angled triangle has a side of 25 opposite an angle of 66°. Find the side adjacent to the angle, to two decimal places.

101. A right-angled triangle has a side of 4.8 opposite an angle of 12°. Find the side adjacent to the angle, to one decimal place.

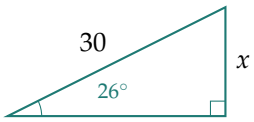
Tier 4 · Top or bottom?

102. In a right-angled triangle, an angle is $44°$. (a) The hypotenuse is 9; find the adjacent side. (b) The adjacent side is 9; find the hypotenuse. Give both to two decimal places.

103. Find x, correct to two decimal places.



104. Find x, correct to two decimal places.



Tier 5 · Practical problems

105. A ladder reaches 4.5 m up a wall and makes an angle of $70°$ with the ground. How long is the ladder, to two decimal places?

106. A path rises 28 m and makes an angle of $6°$ with the horizontal. Find the length of the path, to the nearest metre.

107. A flagpole 9 m tall casts a shadow. The sun's rays make an angle of $55°$ with the ground. Find the length of the shadow, to one decimal place.

Conceptual

108. Explain how you know whether to multiply or divide after writing the trigonometric equation.

109. Liam solves $\sin 30° = \frac{5}{x}$ and gets $x = 5 \sin 30° = 2.5$. Explain his mistake and find x.

110. A right-angled triangle has a side of 10 opposite an angle of $45°$. Without a calculator, find the adjacent side. Explain.

111. Explain why dividing by $\sin\theta$ always gives an answer larger than the side you started with, for an acute angle θ.

112. A kite is flying 32 m above the ground. Its string makes an angle of $46°$ with the ground.

(a) Find the length of the string, to two decimal places. (2 marks)

(b) Find the horizontal distance from the person to the point below the kite. (2 marks)

Revision

113. A 5 m ladder makes an angle of 68° with the ground.

(a) How high up the wall does it reach, to two decimal places? (2 marks)

(b) How far is its foot from the wall, to two decimal places? (2 marks)

(c) Check your answers using Pythagoras' theorem. (1 mark)

114. Two dice are rolled and the numbers are added.

(a) Find $P(\text{total of } 4)$. (1 mark)

(b) Find $P(\text{total is not } 4)$. (1 mark)

(c) The dice are rolled 240 times. How many totals of 4 would you expect? (2 marks)

(d) In 240 rolls, a total of 4 came up 31 times. Find the relative frequency. (1 mark)

115. Liam is paid \$27.60 per hour. Saturday hours are paid at time-and-a-half.

(a) Find his pay for 32 weekday hours. (1 mark)

(b) Find his pay for 6 Saturday hours. (2 marks)

(c) Find his total weekly pay, and his pay for a fortnight. (2 marks)

6 Degrees and minutes

ANCHOR QUESTION

Round 41.2845° *to the nearest minute.*

SYLLABUS MST-12-S1-04

- Apply trigonometric ratios to find the lengths of unknown sides in right-angled triangles given an angle in both degrees, and degrees and minutes

NOTES

- Angles can be measured in degrees and minutes, like hours and minutes: $1^\circ = 60'$.
- $35^\circ 30'$ means 35 degrees and 30 minutes, which is 35.5°.
- Decimal part to minutes: multiply by 60. Minutes to a decimal: divide by 60.
- Round to the nearest minute by rounding the minutes. If they round to $60'$, add 1°.
- On the calculator, the **DMS** key (marked $^\circ\ '\ ''$) enters and converts degrees and minutes. Enter the angle this way inside sin, cos or tan.

WORKED EXAMPLE

Round 56.3726° *to the nearest minute.*

$0.3726 \times 60 = 22.356'$ — decimal part $\times 60$
$\approx 22'$ — round the minutes
$56^\circ 22'$

Tier 1 · Converting

116. Write 48.5° in degrees and minutes.

117. Write 72.4° in degrees and minutes.

118. Write $33^\circ 45'$ as a decimal number of degrees.

Tier 2 · Rounding to the nearest minute

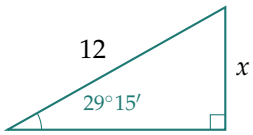
119. Round 19.8462° to the nearest minute.

120. Round 64.1032° to the nearest minute.

121. Round 27.9958° to the nearest minute.

Tier 3 · Sides with degrees and minutes

122. Find x, correct to two decimal places.



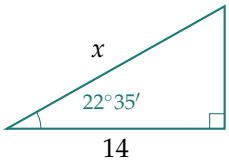
123. A right-angled triangle has a hypotenuse of 20 and an angle of $63°50′$. Find the side adjacent to the angle, to two decimal places.

124. A right-angled triangle has a side of 7 adjacent to an angle of $41°20′$. Find the side opposite the angle, to two decimal places.

Tier 4 · Unknown on the bottom

125. A right-angled triangle has a side of 9 opposite an angle of $36°40′$. Find the hypotenuse, to two decimal places.

126. Find x, correct to two decimal places.



127. A right-angled triangle has a side of 30 opposite an angle of $58°12′$. Find the side adjacent to the angle, to two decimal places.

Tier 5 · Practical problems

128. A 9 m ladder makes an angle of $72°30′$ with the ground. How high up the wall does it reach, to two decimal places?

129. A ramp rises 0.8 m and makes an angle of $5°10′$ with the ground. Find the length of the ramp, to two decimal places.

130. A building casts a shadow 26 m long. The sun's rays make an angle of $47°35′$ with the ground. Find the height of the building, to the nearest metre.

Conceptual

131. Explain why $25.5°$ is $25°30'$ and not $25°50'$.

132. Kim writes $18.9917°$ to the nearest minute as $18°60'$. Explain why this should be written differently.

133. Which angle is larger: $40°45'$ or $40.7°$? Explain.

134. Explain how rounding an angle to the nearest minute is like rounding a time to the nearest minute.

135. A wheelchair ramp rises 0.75 m and makes an angle of $4°30'$ with the ground.

(a) Find the length of the ramp, to two decimal places. (2 marks)

(b) Find the horizontal distance it covers, to two decimal places. (2 marks)

Revision

136. A flight leaves Sydney at 7:35 am and takes 2 hours 55 minutes to reach Auckland, which is 2 hours ahead of Sydney.

(a) Find the arrival time in Sydney time. (1 mark)

(b) Find the local time in Auckland when the flight lands. (1 mark)

(c) The flight is 2160 km. Find the average speed, to the nearest km/h. (2 marks)

(d) Write 2 hours 55 minutes as a decimal, to two decimal places. (1 mark)

137. A right-angled triangle has a hypotenuse of 26 and a shorter side of 10.

(a) Find the third side. (2 marks)

(b) 10 is opposite θ. Write $\sin\theta$ and $\tan\theta$ in simplest form. (2 marks)

(c) Find the perimeter of the triangle. (1 mark)

138. A spinner has 8 equal sections: 3 red, 2 blue and 3 green.

(a) List the sample space. (1 mark)

(b) Find $P(\text{not blue})$. (1 mark)

(c) The spinner is spun twice. Find $P(\text{both blue})$ using an array. (2 marks)

(d) In 120 spins, how many reds would you expect? (1 mark)

Solutions

1. $\sqrt{9+16} = \sqrt{25} = 5$

2. $\sqrt{25+144} = \sqrt{169} = 13$

3. $\sqrt{49+576} = \sqrt{625} = 25$

4. $\sqrt{16+81} = \sqrt{97} = 9.848\ldots \approx 9.85$

5. $\sqrt{6.25+36} = \sqrt{42.25} = 6.5$

6. $\sqrt{121+196} = \sqrt{317} = 17.80\ldots \approx 17.8$

7. $\sqrt{100-64} = \sqrt{36} = 6$

8. $\sqrt{625-225} = \sqrt{400} = 20$

9. $\sqrt{289-64} = \sqrt{225} = 15$

10. $\sqrt{144-49} = \sqrt{95} = 9.746\ldots \approx 9.75$

11. $\sqrt{72.25-17.64} = \sqrt{54.61} = 7.389\ldots \approx 7.39$

12. $\sqrt{400-182.25} = \sqrt{217.75} = 14.756\ldots \approx 14.76$

13. 0.6 m and 0.8 m. $\sqrt{0.36+0.64} = 1$ m

14. $\sqrt{6.76-1} = \sqrt{5.76} = 2.4$ m

15. 1.5 km and 0.8 km. $\sqrt{2.25+0.64} = \sqrt{2.89} = 1.7$ km

16. $AD = \sqrt{169-25} = 12$. $AC = \sqrt{144+256} = \sqrt{400} = 20$

17. $AD = \sqrt{100-36} = 8$. $AC = \sqrt{64+36} = 10$

18. $\sqrt{49+49} = \sqrt{98} = 9.899\ldots \approx 9.90$ cm

19. $c^2 = a^2 + b^2$, so c^2 is bigger than a^2 and bigger than b^2. That means c is longer than both a and b.

20. She added instead of subtracting. A shorter side cannot be longer than the hypotenuse. $\sqrt{100-36} = 8$.

21. $s^2 + s^2 = 100$, so $s^2 = 50$ and $s = \sqrt{50} = 7.071\ldots \approx 7.07$ cm.

22. The theorem adds and subtracts the squares of the lengths, which only makes sense when they are measured in the same unit.

23. (a) $\sqrt{81+144} = \sqrt{225} = 15$ cm. (b) Arc $= \frac{1}{2} \times \pi \times 15 = 23.56\ldots$ cm. Perimeter $= 9 + 12 + 23.56 = 44.56 \approx 45$ cm.

24. (a) $5 \times 2 = 10$. (b) H2, H4: $\frac{2}{10} = \frac{1}{5}$. (c) $1 - \frac{1}{5} = \frac{4}{5}$. (d) $300 \times \frac{1}{5} = 60$.

25. (a) $\frac{4}{2} \times (6+9) = 30$ m^2. (b) $30 \times 0.1 = 3$ m^3. (c) $3 \times \$65 + \$45 = \$240$.

26. (a) $9000 \div 15 = 600$ min $= 10$ h. (b) $15 \times 60 = 900$ L/h. (c) 9000 L $= 9$ kL. $9 \times \$3.20 = \$28.80$.

27. $\sqrt{12.25 - 1} = \sqrt{11.25} = 3.354\ldots \approx 3.35$ m

28. $\sqrt{16 + 1.44} = \sqrt{17.44} = 4.176\ldots \approx 4.18$ m

29. $\sqrt{100 - 64} = 6$ m

30. $\sqrt{225 + 49} = \sqrt{274} = 16.55\ldots \approx 16.6$ cm

31. $\sqrt{3.24 + 1.44} = \sqrt{4.68} = 2.163\ldots \approx 2.16$ m

32. $\sqrt{4225 - 3249} = \sqrt{976} = 31.2\ldots \approx 31$ inches

33. $\sqrt{360\,000 + 640\,000} = \sqrt{1\,000\,000} = 1000$ m

34. $\sqrt{144 + 25} = 13$ km

35. $\sqrt{52\,900 + 9025} = \sqrt{61\,925} = 248.8\ldots \approx 249$ km *cf. 2024 Std 1 HSC Q14(a)*

36. Two sides $= 170$ m. Diagonal $= \sqrt{14\,400 + 2500} = 130$ m. Saving $= 40$ m.

37. Two sides $= 40$ m. Diagonal $= \sqrt{800} = 28.28\ldots$ m. Saving ≈ 11.7 m.

38. Length $= \sqrt{36 + 0.25} = 6.020\ldots$ m. Cost $= 6.020\ldots \times 18 = \$108.37 \approx \$108$.

39. The ladder (hypotenuse) stays the same length. A longer base means $h^2 = c^2 - b^2$ is smaller, so it reaches lower.

40. North and north-east are not at right angles, so the path does not make a right-angled triangle.

41. $\sqrt{16 - 15.21} = \sqrt{0.79} = 0.888\ldots \approx 0.89$ m

42. The diagonal is a straight line between two corners; walking along two sides is a longer path between the same points.

43. (a) $15 \times 400 = 6000$ cm $= 60$ m. (b) $AC = \sqrt{60^2 + 60^2} = \sqrt{7200} = 84.85\ldots \approx 84.9$ m. *cf. 2025 Std 1 HSC Q20*

44. (a) $\frac{6}{20} = \frac{3}{10}$. (b) $1 - \frac{10}{20} = \frac{1}{2} = 50\%$. (c) $60 \times \frac{4}{20} = 12$.

45. (a) $\pi \times 4^2 \times 12 = 603.18\ldots \approx 603$ cm^3. (b) 603 mL. (c) $2 \times \pi \times 4 \times 12 = 301.59\ldots \approx 302$ cm^2.

46. (a) $9.6 \times 50 = 480$ cm $= 4.8$ m. (b) $4.8 \times 2.4 = 11.52$ m^2. (c) Two coats $= 23.04$ m^2. $23.04 \div 12 = 1.92$ L, so one 4 L tin.

47. H $= p$, O $= q$, A $= r$

48. H $= p$, O $= r$, A $= q$

49. YZ (across from X)

50. $\frac{6}{8} = \frac{3}{4}$

51. $\frac{12}{13}$

52. O = 8 (bottom side), H = 17: $\frac{8}{17}$

53. $\frac{3}{5}$, $\frac{4}{5}$, $\frac{3}{4}$

54. O = 24: $\frac{24}{25}$, $\frac{7}{25}$, $\frac{24}{7}$

55. $\frac{20}{29}$, $\frac{21}{29}$, $\frac{20}{21}$

56. O and H: $\sin 40^\circ = \frac{x}{20}$

57. O and A: $\tan 30^\circ = \frac{x}{15}$

58. A and H: $\cos 52^\circ = \frac{x}{9}$

59. 0.7660

60. 0.3090

61. $\tan 25^\circ = 0.4663$, $\tan 65^\circ = 2.1445$. $\tan 65^\circ$ is greater than 1.

62. Both have the hypotenuse on the bottom, and the hypotenuse is the longest side, so the fraction is less than 1.

63. Tan is opposite over adjacent: $\tan\theta = \frac{O}{A}$ (TOA). He has the fraction upside down.

64. O = 3, H = 5, so A = $\sqrt{25-9} = 4$. $\cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4}$.

65. The opposite side is the one across from the marked angle, so a different angle has a different side across from it. The hypotenuse stays the same.

66. (a) $\sqrt{1681-81} = \sqrt{1600} = 40$. (b) $\tan\theta = \frac{9}{40}$. (c) $\cos\theta$.

67. (a) $\sqrt{49-3.24} = \sqrt{45.76} = 6.764\ldots \approx 6.76$ m. (b) $4 \times 1.8 = 7.2$ m. The height is only 6.76 m, so no. (c) $1.8 : 6.76 \approx 1 : 3.76$.

68. (a) HA, HB, HC, TA, TB, TC. (b) $\frac{1}{6}$. (c) HA, HB, HC, TB: $\frac{4}{6} = \frac{2}{3}$.

69. (a) $0.02 \times \$1240 = \$24.80$. (b) $\$1240 - \$214 - \$24.80 = \$1001.20$. (c) $52 \times \$1001.20 = \$52\,062.40$.

70. $x = 8\sin 30^\circ = 4$

71. $15\sin 42^\circ = 10.036\ldots \approx 10.04$

72. $6.4\sin 67^\circ = 5.891\ldots \approx 5.89$

73. $x = 11\cos 38^\circ = 8.668\ldots \approx 8.67$

74. $24\cos 55^\circ = 13.765\ldots \approx 13.77$

75. From 22° at the top, x is adjacent. $x = 9.5\cos 22^\circ = 8.808\ldots \approx 8.81$

76. $x = 12\tan 33^\circ = 7.792\ldots \approx 7.79$

77. $5\tan 61^\circ = 9.020\ldots \approx 9.02$

78. $40\tan 15^\circ = 10.71\ldots \approx 10.7$

79. From 36° at the top: 14 is A, x is O. $x = 14\tan 36^\circ = 10.171\ldots \approx 10.17$

80. H and A: $x = 18\cos 27^\circ = 16.038\ldots \approx 16.04$

81. From 48° at the top: x is O, 7.5 is H. $x = 7.5\sin 48^\circ = 5.573\ldots \approx 5.57$

82. $7\sin 75^\circ = 6.761\ldots \approx 6.76$ m

83. $50\sin 40^\circ = 32.1\ldots \approx 32$ m

84. $18\tan 34^\circ = 12.14\ldots \approx 12.1$ m

85. Label the sides from the angle. Use the ratio that contains the side you know and the side you want: O and H sine, A and H cosine, O and A tangent.

86. Her calculator is in radian mode. In degree mode $10\sin 30^\circ = 5$. A length cannot be negative.

87. $10\sin 60^\circ = 8.66$ and $10\sin 30^\circ = 5$. The larger angle is opposite the longer side.

88. With a 45° angle, the other acute angle is also 45°, so the opposite and adjacent sides are equal and $\frac{O}{A} = 1$.

89. (a) $6.2\sin 6^\circ = 0.648\ldots \approx 0.65$ m. (b) $6.2\cos 6^\circ = 6.166\ldots \approx 6.17$ m.

90. (a) $\sqrt{105^2 + 68^2} = \sqrt{15\,649} = 125.09\ldots \approx 125$ m. (b) $105 \times 68 = 7140$ m^2 $= 0.714$ ha. (c) $3 \times 2 \times (105 + 68) = 1038$ m $= 1.038$ km.

91. (a) $\frac{18}{75} = 0.24$. (b) $1200 \times 0.24 = 288$. (c) $288 \times \$5.20 = \$1497.60$.

92. (a) 3:30 am. (b) 3:30 am +3 h = 6:30 am. (c) 4 h 10 min $= 4.1\dot{6}$ h. $3290 \div 4.1\dot{6} = 789.6 \approx 790$ km/h.

93. $x = \frac{6}{\sin 28^\circ} = 12.780\ldots \approx 12.78$

94. $\frac{15}{\sin 62^\circ} = 16.988\ldots \approx 16.99$

95. $\frac{3.6}{\sin 19^\circ} = 11.057\ldots \approx 11.06$

96. $x = \frac{20}{\cos 32^\circ} = 23.583\ldots \approx 23.58$

97. $\frac{8}{\cos 71^\circ} = 24.572\ldots \approx 24.57$

98. From 24° at the top, 13 is adjacent. $x = \frac{13}{\cos 24^\circ} = 14.23\ldots \approx 14.2$

99. $\tan 37^\circ = \frac{10}{x}$, $x = \frac{10}{\tan 37^\circ} = 13.270\ldots \approx 13.27$

100. $\frac{25}{\tan 66^\circ} = 11.130\ldots \approx 11.13$

101. $\frac{4.8}{\tan 12^\circ} = 22.58\ldots \approx 22.6$

102. (a) $9\cos 44^\circ = 6.47$ (top). (b) $\frac{9}{\cos 44^\circ} = 12.51$ (bottom).

103. From 52° at the top: 16 is O, x is A. $\tan 52^\circ = \frac{16}{x}$, $x = \frac{16}{\tan 52^\circ} = 12.500\ldots \approx 12.50$

104. x is on top: $x = 30\sin 26^\circ = 13.151\ldots \approx 13.15$

105. $\frac{4.5}{\sin 70^\circ} = 4.788\ldots \approx 4.79$ m

106. $\frac{28}{\sin 6^\circ} = 267.8\ldots \approx 268$ m *cf. 2024 Std 1 HSC Q25*

107. $\frac{9}{\tan 55^\circ} = 6.30\ldots \approx 6.3$ m

108. If x is on top of the fraction, multiply by the bottom number. If x is on the bottom, divide the known side by the ratio.

109. x is on the bottom, so he should divide: $x = \frac{5}{\sin 30^\circ} = 10$. His answer is also shorter than the opposite side, which is impossible for a hypotenuse.

110. 10. With a 45° angle, the triangle is isosceles, so the opposite and adjacent sides are equal.

111. For an acute angle, $\sin\theta$ is less than 1, and dividing by a number less than 1 makes the result larger.

112. (a) $\frac{32}{\sin 46^\circ} = 44.485\ldots \approx 44.49$ m. (b) $\frac{32}{\tan 46^\circ} = 30.902\ldots \approx 30.90$ m.

113. (a) $5\sin 68^\circ = 4.635\ldots \approx 4.64$ m. (b) $5\cos 68^\circ = 1.873\ldots \approx 1.87$ m. (c) $\sqrt{4.635^2 + 1.873^2} = 5.00$ m. ✓

114. (a) $(1,3), (2,2), (3,1)$: $\frac{3}{36} = \frac{1}{12}$. (b) $\frac{11}{12}$. (c) $240 \times \frac{1}{12} = 20$. (d) $\frac{31}{240} = 0.129\ldots \approx 0.13$.

115. (a) $32 \times \$27.60 = \$883.20$. (b) $6 \times 1.5 \times \$27.60 = \$248.40$. (c) $\$883.20 + \$248.40 = \$1131.60$ a week; \$2263.20 a fortnight.

116. $48^\circ 30'$

117. $0.4 \times 60 = 24$: $72^\circ 24'$

118. $45 \div 60 = 0.75$: 33.75°

119. $0.8462 \times 60 = 50.77'$: $19^\circ 51'$

120. $0.1032 \times 60 = 6.19'$: $64^\circ 6'$

121. $0.9958 \times 60 = 59.75'$, which rounds to $60'$: $28^\circ 0'$

122. $x = 12\sin 29^\circ 15' = 5.862\ldots \approx 5.86$

123. $20\cos 63^\circ 50' = 8.817\ldots \approx 8.82$

124. $7\tan 41^\circ 20' = 6.157\ldots \approx 6.16$

125. $\frac{9}{\sin 36^\circ 40'} = 15.074\ldots \approx 15.07$

126. $x = \frac{14}{\cos 22^\circ 35'} = 15.164\ldots \approx 15.16$

127. $\frac{30}{\tan 58^\circ 12'} = 18.596\ldots \approx 18.60$

128. $9\sin 72^\circ 30' = 8.583\ldots \approx 8.58$ m

129. $\frac{0.8}{\sin 5°10'} = 8.882\ldots \approx 8.88$ m

130. $26 \tan 47°35' = 28.45\ldots \approx 28$ m

131. There are 60 minutes in a degree, so half a degree is 30 minutes, not 50.

132. $60' = 1°$, so $18°60'$ is $19°0'$. Minutes are always less than 60.

133. $40°45' = 40.75°$, which is larger than $40.7°$.

134. Both use 60 minutes in one unit (a degree or an hour), so the decimal part is multiplied by 60 and then rounded.

135. (a) $\frac{0.75}{\sin 4°30'} = 9.557\ldots \approx 9.56$ m. (b) $\frac{0.75}{\tan 4°30'} = 9.528\ldots \approx 9.53$ m.

136. (a) 10:30 am. (b) 12:30 pm. (c) $2160 \div 2.91\dot{6} = 740.57\ldots \approx 741$ km/h. (d) 2.92 h.

137. (a) $\sqrt{676 - 100} = 24$. (b) $\sin\theta = \frac{10}{26} = \frac{5}{13}$, $\tan\theta = \frac{10}{24} = \frac{5}{12}$. (c) $26 + 24 + 10 = 60$.

138. (a) $S = \{R, B, G\}$. (b) $1 - \frac{2}{8} = \frac{3}{4}$. (c) $8 \times 8 = 64$ cells, $2 \times 2 = 4$ both blue: $\frac{4}{64} = \frac{1}{16}$. (d) $120 \times \frac{3}{8} = 45$.